

**Dear Oxford Economics Hiring Team,**

I am pursuing the Spatial Data Scientist position in Oxford Economics' Cities & Regions team. I have five years of experience in urban design and spatial planning, along with graduate training in computational urban analysis at CASA, UCL. I am especially interested in Oxford Economics because of your work linking detailed spatial data to sub-national economic outcomes, such as the Global Cities Index and the advanced geospatial methods your Cities team is developing.

For my MSc dissertation, I worked with CACI Ltd through the GeoDS industry placement scheme to examine whether the type of urban fabric can predict retail spending patterns in UK cities, independent of demographic factors. Using CACI's consumer spending data and HiMoC classifications, I am building a Python pipeline with GeoPandas, scikit-learn, and spatial autocorrelation analysis to model how urban form relates to economic behavior throughout the country. The main challenge is to separate the effect of urban form on spending while taking into account demographic differences at the LSOA level, which I feel is similar to the spatial economic modeling your team does. This project has involved handling large geospatial datasets, making methodological choices for a commercial partner, and sharing results with both technical and non-technical audiences.

Before my MSc, I led a team of four architects at Derman Verbakel Architecture, managing fifteen projects for municipal clients. These included a transportation masterplan for Eilat Municipality and a TOD regeneration plan for 3,000 units. I also developed geocomputational workflows in QGIS and ArcGIS for mapping and market analysis, which gave me hands-on experience turning spatial data into outcomes for city planning and, by extension, all that is related to supporting the city.

I have strong Python skills with GeoPandas, scikit-learn, and spatial network analysis, as well as experience with reproducible, version-controlled workflows. I also have a record of explaining complex geographical results to non-specialists. I would be excited to contribute to Oxford Economics' work in location intelligence and sub-national forecasting.

Thank you for considering my application.

**Warm regards,**

**Emily Deeb**

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